Predicting Store-Level Liquor Purchases from Transaction History



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Data: Iowa Liquor Sales 2026 (Iowa Department of Revenue, CC BY)

Distributors commit inventory before they know what accounts will order

Two costly ways to miss

Order too much
Working capital sits in slow-moving stock and warehouse space

Order too little
A fast mover runs short: lost revenue and lost account confidence

Root cause: the distributor does not know what each account will order next.

Research questions

Q1 Can purchase history predict a store's next four weeks of purchases better than a naive carry-forward?

Q2 Which product, store, and timing characteristics matter most for the forecast?

Q3 Is forecasting more dependable for a product within a store, or a category within a store?

The data: every spirits purchase by Iowa retailers, January to August 2026

1.60M

invoice lines after cleaning

2,203

stores across all 99 Iowa counties

4,429

products in 44 categories

19.5M

net bottles purchased

Why Iowa?

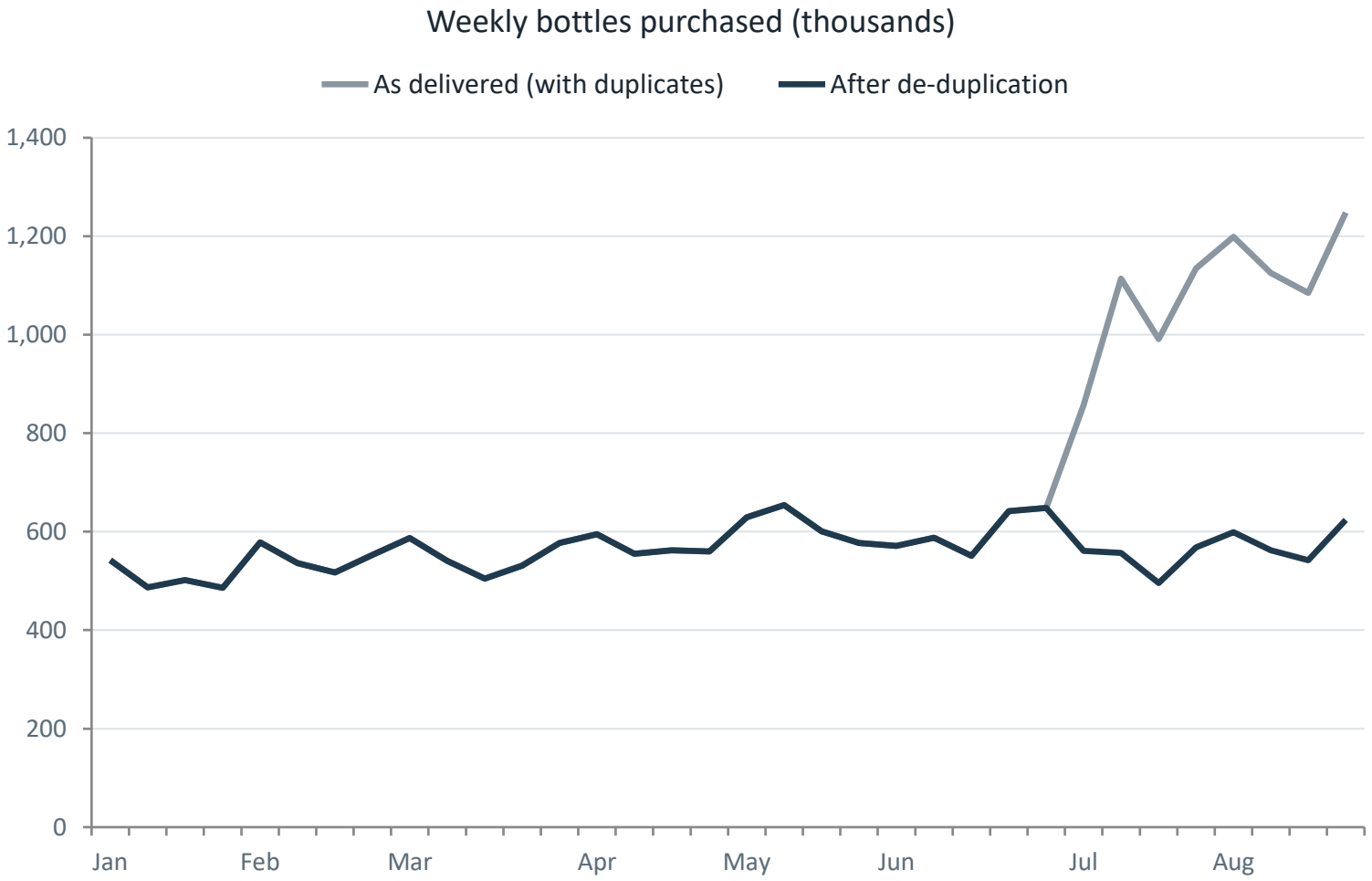
Iowa is an alcohol control state: the state is the only wholesaler of spirits, so every purchase by a grocery, liquor, or convenience store is published at the invoice-line level.

What it is not

These are retailer purchases, not inventory on hand, stockouts, or consumer sales. The models predict ordering behavior only.

Source: Iowa Department of Revenue (2026), Iowa Liquor Sales 2026, dataset 1263, Iowa Data Hub (CC BY).

Cleaning caught an export artifact that doubled July and August volume

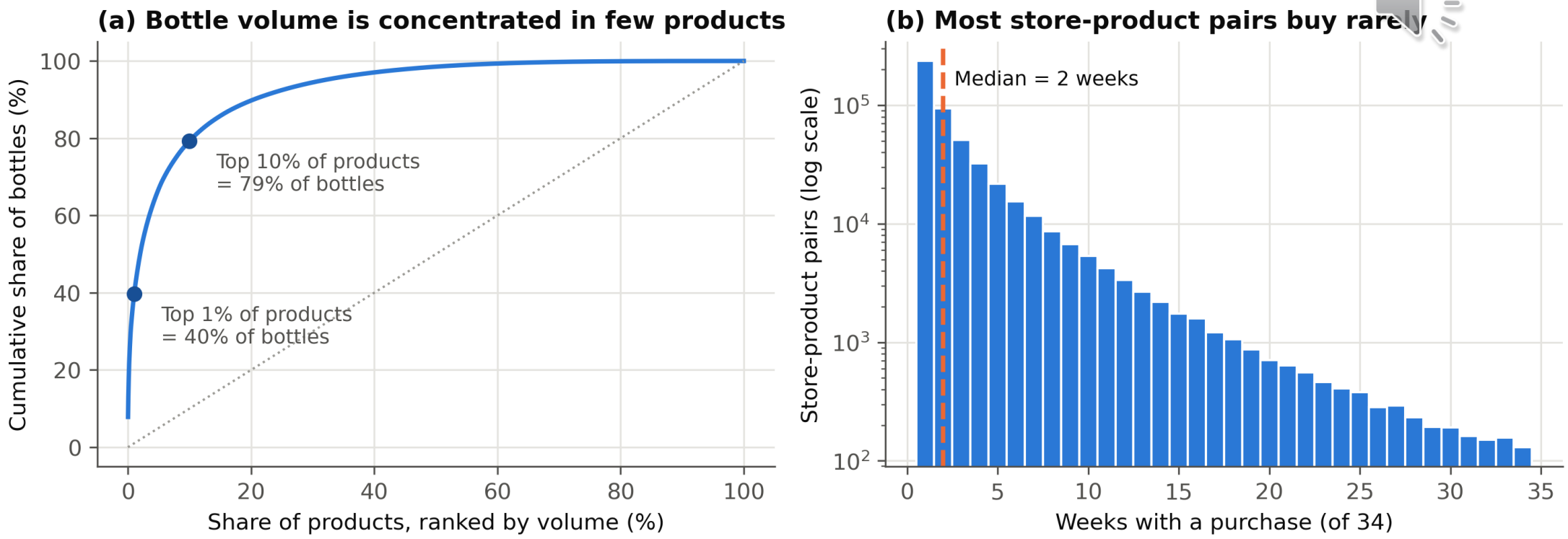


364,160

exact duplicate rows removed, almost all in July (43% of rows) and August (50%)

- Clean count matched the state portal to within 14 rows
- Missing store geography filled from the same store
- Returns netted into weekly totals
- Summed to store × product × week over 34 full weeks

Demand is concentrated in few products and most pairs buy rarely



Eligibility rule: forecast a series only if it bought in at least 3 of the previous 8 weeks, judged from past weeks alone. Eligible series cover 37% of product-level bottles and 88% of category-level bottles.

Validation mimics real forecasting: learn from the past, test on unseen weeks



What is predicted

Bottles a store buys in the next four weeks, for each product and each category: a monthly replenishment rhythm.

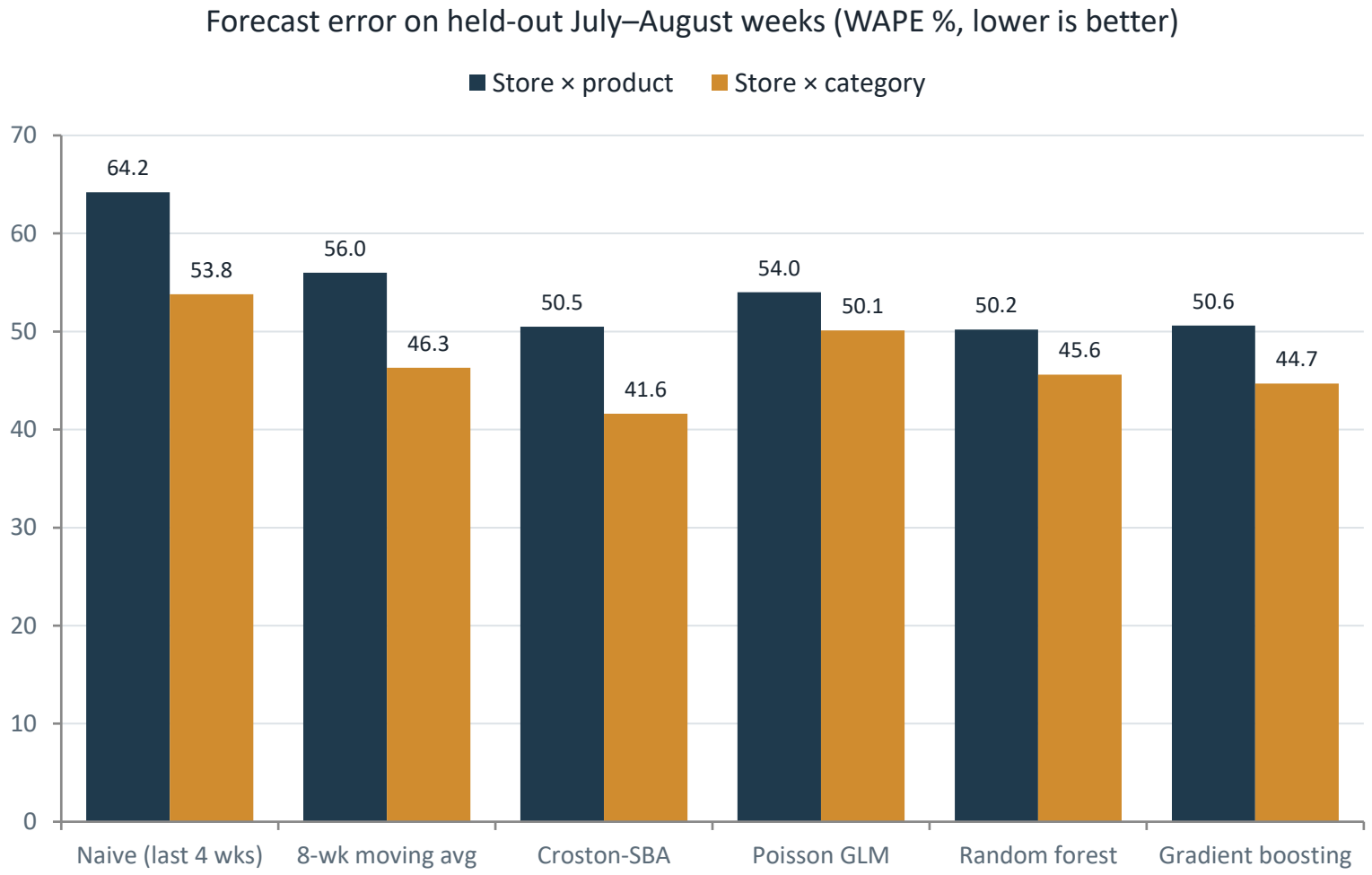
What is compared

3 baselines: naive last 4 weeks, 8-week average, Croston-SBA.
3 models: Poisson regression, random forest, gradient boosting.

How it is scored

WAPE = total error ÷ total actual bottles.
Rolling cross-validation picked the model; the test set was used once.

Q1: History beats naive by about 21%, but a simple baseline matched ML

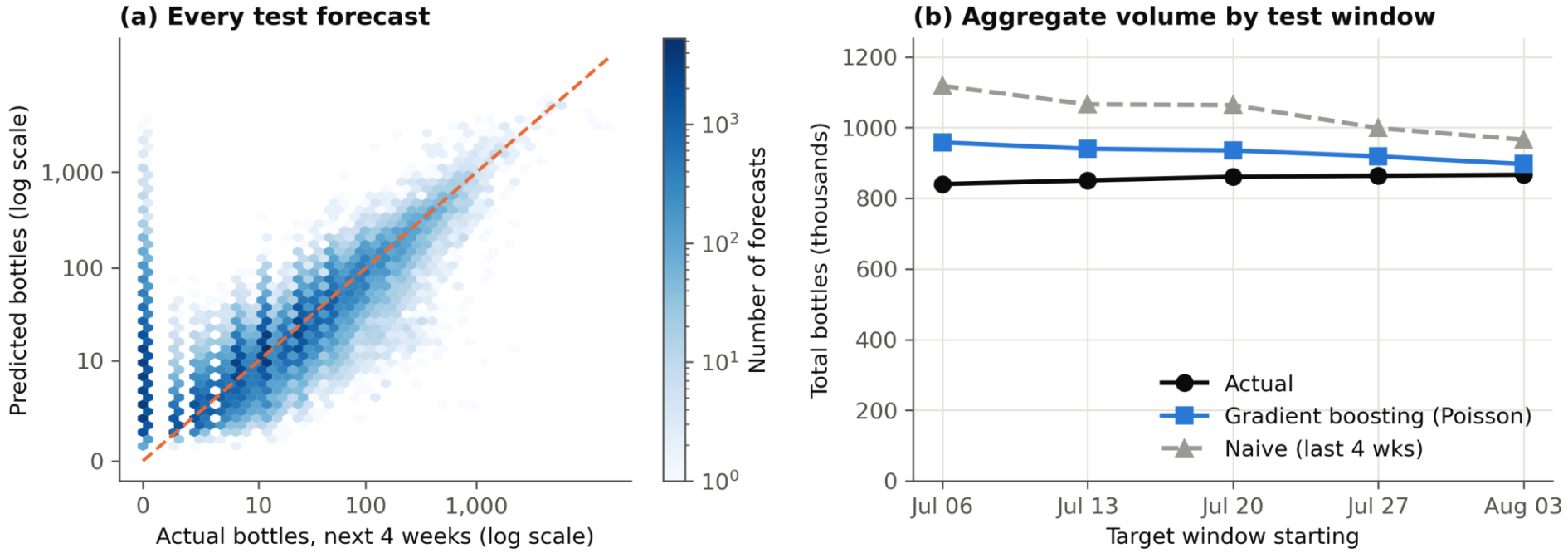


21%

less error than carrying last month forward (store × product, gradient boosting)

- Every model and smoothed baseline beat naive: H1 supported
- Croston-SBA tied the ML models (50.5% vs 50.2–50.6%) and won at category level
- Gradient boosting's cross-validation lead did not hold in July–August

Forecasts track regular buyers closely and over-forecast far less than naive

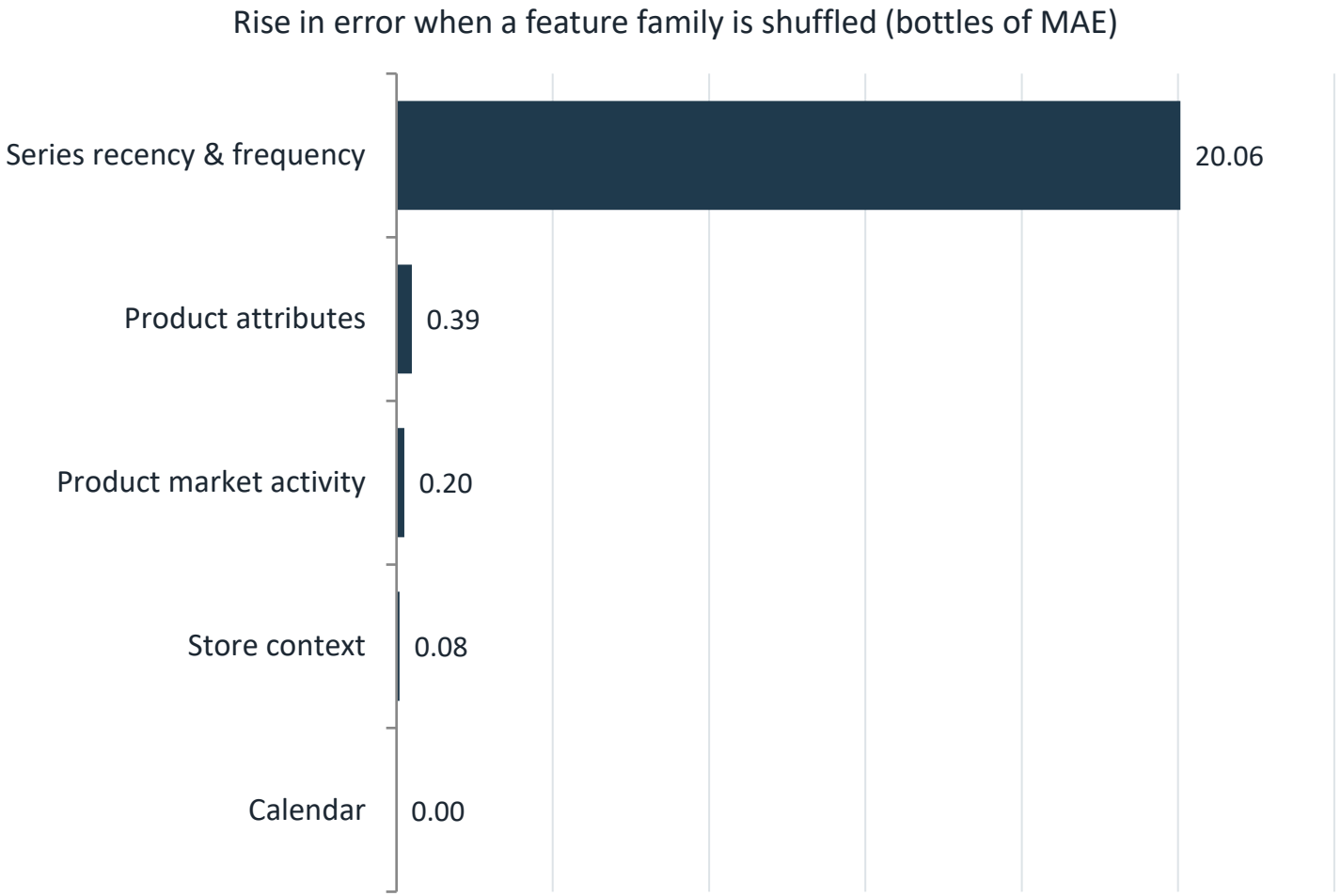


59% of individual forecasts closer to actual than naive

+8.6% aggregate over-forecast, versus +21.7% for naive

22.8% error on store-level four-week totals (naive 32.5%)

Q2: A store's own recent buying pattern drives the forecast



Strongest individual features

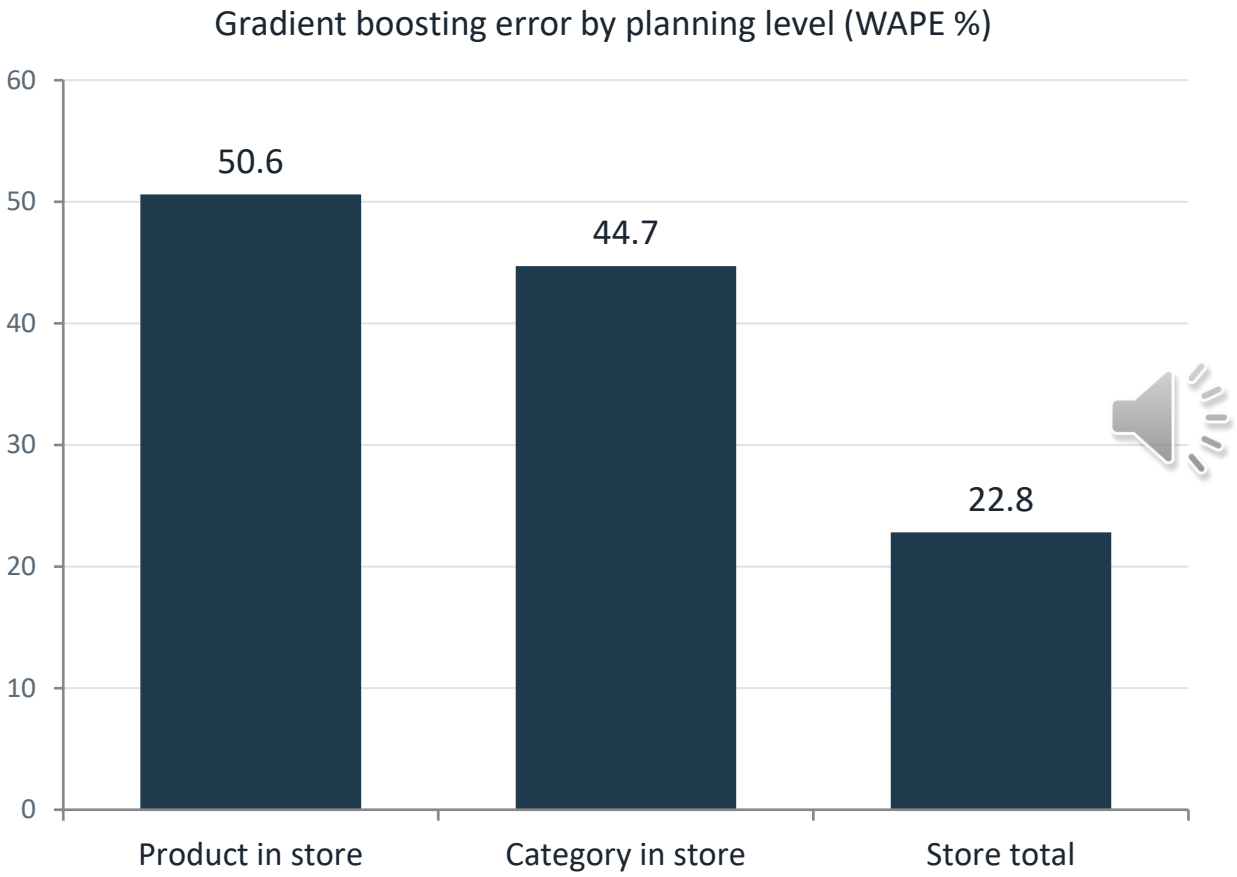
+16.4 Eight-week purchase total

+3.3 Active weeks in the last eight

+1.5 Long-run share of weeks with a purchase

H2 supported. Importance shows association, not cause; eight months is too short to test seasonality.

Q3: Coarser planning levels are far more dependable



Best category-level method, Croston-SBA, reached 41.6%. Coverage rises from 37% to 88% of bottles at category level, but the margin over naive does not grow.

Where error is highest (product in store)

Low-volume items
78.5% WAPE vs 46.9% for the busiest quartile

Smallest stores
56.3% WAPE vs 48–51% for larger stores

Rural counties
54.8% WAPE vs 46.0% in dense counties

The model still beat naive in every segment.

Ethical risks were identified, assessed, and mitigated

Business re-identification
Likelihood high · impact moderate

Every store is named in the public data. No store names in any figure; examples anonymized.

Geographic and size bias
Observed · impact moderate

Small and rural stores see higher error. Results reported by segment; extra planner review there.

Data-quality distortion
Certain · impact high

Duplicates, rounding, returns. Duplicates removed and reconciled; every cleaning rule logged.

Over-reliance on forecasts
Moderate · impact high

Purchases are not inventory and the model over-forecasts. Forecasts advisory; a person places every order.

Public CC BY data, no personal information, verified against the portal. Commercial use would need data-sharing agreements and trade-practice review.

Limitations: what these results cannot tell us yet

No full seasonal cycle

Eight months of data; the test covers only July and August, so holiday accuracy is unknown.

Purchases, not inventory

Nothing here observes stock on hand, so the models cannot predict stockouts.

One control state

Iowa's single wholesaler and uniform prices may not transfer to open markets.

Partial coverage

Eligible series cover 37% of product-level volume; models lightly tuned, no prediction intervals.

Recommendations and a phased path to implementation

What a distributor should do

1. Replace carry-forward planning with a smoothed, intermittency-aware baseline such as Croston-SBA
2. Plan volume at the account and category level; use item forecasts as review prompts
3. Give extra attention to low-volume items, small stores, and rural accounts
4. Hold off on operational machine learning until it beats the baseline out of sample

Rollout in a distributor platform (Pourtion)

Phase 1	Croston-SBA forecasts rolled up to account and category totals; flag big departures from recent ordering
Phase 2	Gradient boosting in shadow mode; promote only if it beats the baseline for 3+ months including holidays
Phase 3	Add inventory and lead times to suggest reorder points

A person reviews and places every order.

Answers to the research questions

Q1

Yes, history beats naive

About 21% less error than carrying last month forward, but simple Croston-SBA matched machine learning.

Q2

Recent buying drives it

A series' own eight-week volume and purchase frequency dominate; product, store, and calendar add little.

Q3

Plan at coarser levels

Category (42%) and store totals (23%) are planning-ready; item level (~50%) is best used as a review prompt.

Thank you. White paper, code, and audience Q&A accompany this presentation.